

1. $\lceil \log_2 51 \rceil = 6$ 需要6位, 最多表示 $2^6 = 64$ 名学生

表示51: $(00110011)_2$ $(33)_{16}$ $(01010001)_{BCD}$

-29补码: 11100011

BCD码竖式计算 $51 + 29$:

$$\begin{array}{r} 51 = 0101\ 0001 \\ + 29 = 0010\ 1001 \\ \hline 0111\ 1010 \quad \text{后四位溢出} \\ + \quad 0110 \\ \hline \end{array}$$

3. 1) $X+1 = (X+1) \cdot 1$ 恒等性

$$\begin{aligned} &= (X+1) \cdot (X+X') \quad \text{互补} \\ &= X + (1 \cdot X') \quad \text{分配律} \\ &= X + (X' \cdot 1) \quad \text{交换律} \\ &= X + X' \quad \text{恒等性} \\ &= 1 \quad \text{互补} \end{aligned}$$

2) $X + XY = X \cdot 1 + X \cdot Y$ 恒等性

$$\begin{aligned} &= X \cdot (1+Y) \quad \text{分配律} \\ &= X \cdot (Y+1) \quad \text{交换律} \\ &= X \cdot 1 \quad \text{空单元} \\ &= X \quad \text{恒等性} \end{aligned}$$

4. 中间10编码为: $00|0, 01|0, 01|1, 01|0, 01|00, 11|00, 11|01, 11|1, 11|10, 10|10$
 分别代表: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
 故真值表为:

A	B	C	D	$f(A,B,C,D)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1